FRAUD DETECTION IN TRANSACTIONS

Course: Data Analytics

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ABSTRACT

With the rapid growth of digital banking and online transactions, detecting fraudulent activities has become a major challenge for financial institutions. This project focuses on analyzing transaction data to identify suspicious patterns and classify transactions as fraudulent or genuine using data analytics and machine learning techniques.

The dataset includes transaction details such as transaction amount, date, type, location, login attempts, and account balance. Data preprocessing and feature engineering techniques were applied to create useful features like transaction hour and amount range. Exploratory Data Analysis (EDA) was performed to understand transaction behaviour and identify abnormal patterns.

Machine learning techniques were applied to detect fraudulent transactions using the Logistic Regression model. This model was used to classify transactions as fraudulent or genuine based on the given features.

The performance of the model was evaluated using key metrics such as accuracy, confusion matrix, precision, and recall, which helped in understanding how effectively the model identifies fraudulent transactions.

The results demonstrate that data-driven approaches can effectively detect fraud patterns and assist financial institutions in improving security, minimizing financial losses, and enhancing transaction monitoring systems.

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1. INTRODUCTION

Digital transactions have increased significantly with the advancement of online banking, e-commerce, and mobile payment systems. While these technologies provide convenience and speed, they also introduce new risks, particularly in the form of fraudulent activities. As the volume of transactions continues to grow, identifying and preventing fraud has become a major challenge for financial institutions. This project focuses on analyzing transaction data and detecting fraud using machine learning techniques. By identifying unusual patterns such as high transaction amounts, suspicious login attempts, and abnormal transaction timings, the system helps in detecting fraudulent transactions effectively. In addition to pattern detection, data preprocessing and feature engineering techniques are applied to improve data quality and extract meaningful insights. Features such as transaction hour and amount range help in understanding user behaviour more clearly. Exploratory Data Analysis (EDA) is performed to study transaction trends, detect anomalies, and identify high-risk patterns. These insights play a crucial role in building an effective fraud detection model. A Logistic Regression model is used to classify transactions as fraudulent or genuine. The model is evaluated using performance metrics such as accuracy, precision, recall, and confusion matrix to ensure its effectiveness. Overall, this project aims to provide a data-driven approach to fraud detection, helping organizations reduce financial losses, enhance security, and improve decision-making processes.

- **Project Background**

With the increasing use of digital platforms, financial fraud has become a critical concern. Fraudulent transactions can lead to financial losses and security risks for both customers and institutions. Organizations are now using data analytics and machine learning to detect fraud patterns and prevent unauthorized transactions. This project aims to analyze transaction data and build a system that can identify fraudulent activities.

- **Problem Statement**

Detecting fraudulent transactions is challenging due to large volumes of transaction data and evolving fraud techniques. Traditional methods fail to identify hidden patterns effectively.
This project aims to develop a fraud detection system that analyses transaction data and classifies transactions as fraudulent or normal using machine learning techniques.

❖ **Objectives**

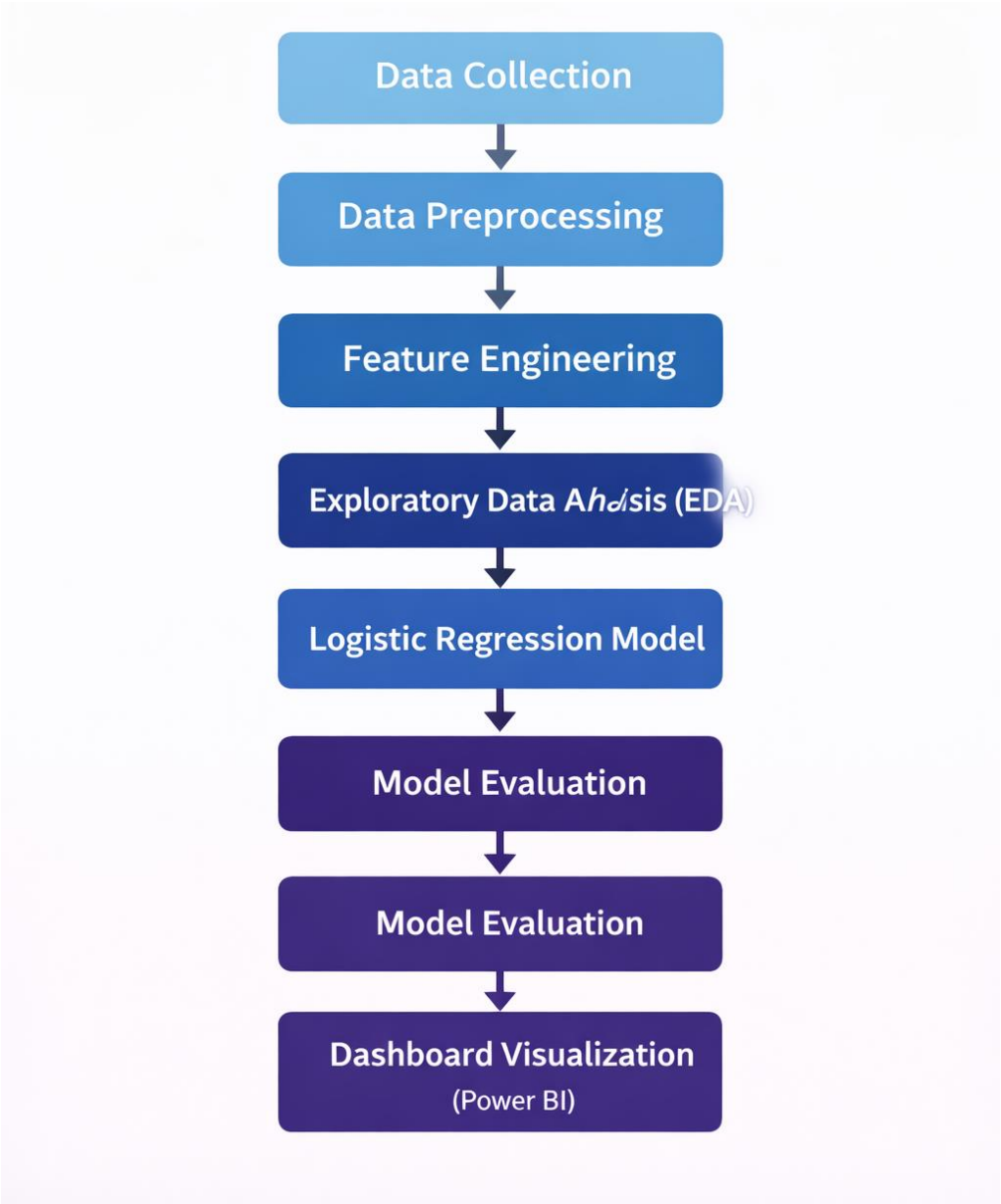
- To analyze transaction data and identify patterns associated with fraudulent activities
- To perform data preprocessing and apply feature engineering techniques for better analysis
- To implement machine learning (Logistic Regression) for fraud detection
- To evaluate the performance of the model using appropriate metrics
- To visualize key insights and patterns through an interactive dashboard

2. TOOLS AND TECHNOLOGIES

This project uses various tools and technologies for data analysis, model development, and visualization.

- **Python** – Used for data analysis, processing, and implementing machine learning models
- **Pandas, NumPy** – Used for data manipulation, cleaning, and numerical computations
- **Scikit-learn** – Used to build and evaluate the Logistic Regression model for fraud detection
- **Matplotlib & Seaborn** – Used for data visualization and exploratory data analysis
- **MySQL** – Used to store and analyze transaction data using SQL queries
- **Power BI** – Used to create interactive dashboards and visualize fraud insights

3. SYSTEM ARCHITECTURE



4. DATA DESCRIPTION

4.1 Dataset Overview

The dataset used in this project consists of transaction records containing various attributes related to financial activities. It includes details such as transaction amount, date and time, transaction type, location, and user behaviour indicators.

4.2 Dataset Features

The dataset contains the following attributes used for analysis and fraud detection:

- **TransactionID** – Unique identifier for each transaction
- **AccountID** – Unique identifier for the customer account
- **TransactionAmount** – Amount involved in the transaction
- **TransactionDate** – Date and time of the transaction
- **TransactionType** – Type of transaction (Online, ATM, POS, etc.)
- **Location** – Location where the transaction occurred
- **DeviceType** – Type of device used for the transaction (if present)
- **IP_Address** – IP address of the user during transaction (if present)
- **LoginAttempts** – Number of login attempts made before the transaction
- **AccountBalance** – Available account balance at the time of transaction
- **Hour** – Extracted feature representing transaction hour
- **Day/Weekday** – Derived feature showing day of transaction (if created)
- **Amount_Range** – Categorized transaction amount (Low, Medium, High)
- **High_Amount** – Binary feature indicating high-value transactions (if created)
- **FraudFlag** – Target variable indicating fraud (0 = Normal, 1 = Fraud)

4.3 Data Usage

The dataset is utilized for performing exploratory data analysis, feature engineering, and building a machine learning model to detect fraudulent transactions. The features help in identifying patterns such as high-value transactions, unusual login activity, and abnormal transaction timings, which are key indicators of fraud.

5. DATA PREPROCESSING AND FEATURE ENGINEERING

5.1 Data Preprocessing

Data preprocessing is performed to prepare the dataset for analysis and modeling. This step ensures that the data is clean, structured, and suitable for further processing.

- The dataset is examined for data types and structure using .info()
- Necessary transformations are applied to ensure proper data formatting
- The dataset is validated to ensure consistency before feature engineering

5.2 Feature Engineering

Feature engineering is carried out to create new meaningful features from existing data, which helps in improving model performance and analysis.

1. Transaction Hour Extraction

- A new feature **Hour** is created by extracting the hour component from the transaction date
- This helps in analyzing transactions based on time and identifying unusual timings

2. Transaction Amount Categorization

- A new feature **Amount_Range** is created by categorizing transaction amounts into three groups Low, Medium,High
- This helps in understanding transaction distribution and identifying high-value transactions

3. Fraud Flag Creation

- A new target variable FraudFlag is created using rule-based conditions
- A transaction is marked as fraud (1) if:

  Transaction amount is high (> 400)

  Transaction occurs during unusual hours (< 5 AM)

  Login attempts exceed normal limit (> 3)

- Otherwise, it is marked as normal (0)

4. High Amount Indicator

- A binary feature High_Amount is created
- It identifies whether a transaction amount exceeds 500
- Helps in highlighting high-risk transactions

5.3 Data Verification

The dataset structure is verified using .info()

Fraud distribution is checked using value counts

This ensures that the newly created features are correctly added and ready for analysis

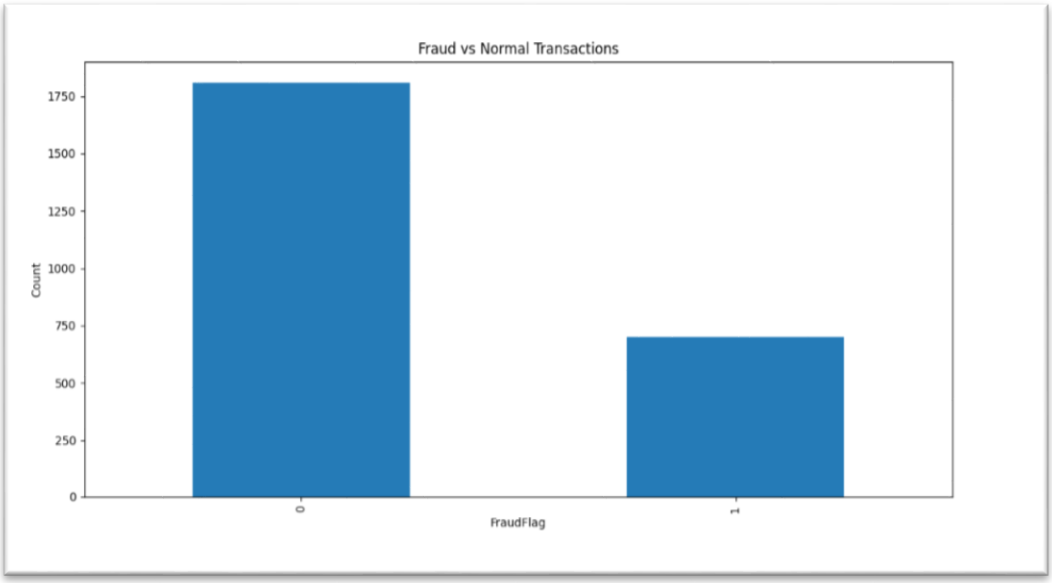
```
PROBLEMS    OUTPUT    TERMINAL    ...                Python: fraud_detection_transaction

      TransactionID AccountID  ...  AccountBalance PreviousTransactionDate
 1    AccountID                2512 non-null    object
 2    TransactionAmount        2512 non-null    float64
 3    TransactionDate          2512 non-null    datetime64[ns]
 4    TransactionType          2512 non-null    object
 5    Location                 2512 non-null    object
 6    DeviceID                 2512 non-null    object
 7    IP Address               2512 non-null    object
 8    MerchantID               2512 non-null    object
 9    Channel                  2512 non-null    object
 10   CustomerAge              2512 non-null    int64
 11   CustomerOccupation       2512 non-null    object
 12   TransactionDuration      2512 non-null    int64
 13   LoginAttempts            2512 non-null    int64
 14   AccountBalance           2512 non-null    float64
 15   PreviousTransactionDate  2512 non-null    object
 16   Hour                     2512 non-null    int32
 17   Amount_Range             730 non-null     category
 18   FraudFlag                2512 non-null    int64
 19   High_Amount              2512 non-null    int64
dtypes: category(1), datetime64[ns](1), float64(2), int32(1), int64(5), object(10)
```

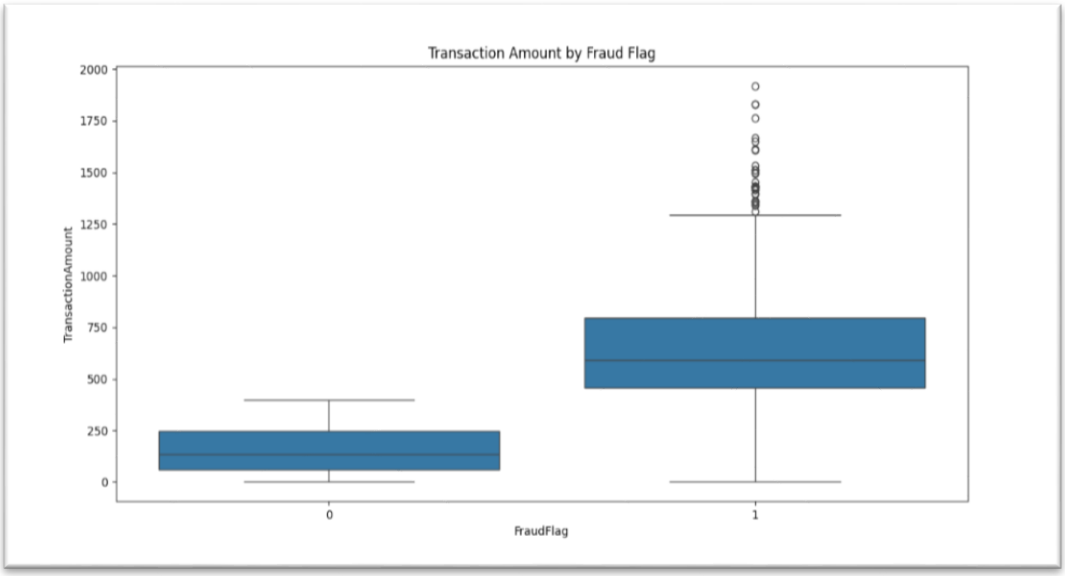
6. EXPLORATORY DATA ANALYSIS (EDA)

➢ **Fraud vs Normal Transaction Analysis**

- The distribution of fraudulent and normal transactions is analyzed using count plots
- A bar chart is used to visualize the number of fraud (1) and normal (0) transactions
- This helps in understanding the imbalance in the dataset and the proportion of fraud cases



- **Transaction Amount Distribution**
  - A boxplot is used to compare transaction amounts between fraud and normal transactions
  - This visualization helps in identifying outliers and high-value transactions
  - It provides insights into whether fraudulent transactions tend to have higher or unusual amounts



- **Average Transaction Amount Analysis**
  - The average transaction amount is calculated for both fraud and normal transactions
  - This helps in understanding the difference in spending patterns between genuine and fraudulent activities
  - Higher average values in fraud transactions may indicate suspicious behavior

- **Risk Analysis**
  - Risk analysis is performed by calculating statistical measures such as:
    - Mean (average transaction amount)
    - Maximum transaction amount
    - Minimum transaction amount
  - This analysis helps in identifying high-risk transaction ranges and understanding the variability in transaction behavior

7. PREDICTIVE MODELLING

In this project, the **Logistic Regression** model is used to classify transactions as fraudulent or normal. Logistic Regression is a supervised machine learning algorithm suitable for binary classification problems, making it appropriate for fraud detection where the output is either fraud (1) or normal (0).

Feature Selection:

The following features are selected as input variables for the model:

- TransactionAmount – Represents the transaction value
- Hour – Indicates the time of transaction
- High_Amount – Identifies high-value transactions

The target variable used is:

- FraudFlag – Indicates whether a transaction is fraudulent (1) or normal (0)

Data Splitting

✓ The dataset is divided into training and testing sets

✓ 70% of the data is used for training and 30% for testing

✓ This helps in evaluating the model's performance on unseen data

Data Scaling

✓ Feature scaling is applied using StandardScaler

✓ This ensures that all input features are on the same scale

✓ It improves the performance of the Logistic Regression model

8. MODEL IMPLEMENTATION

Model Training

- The Logistic Regression model is trained using the training dataset after splitting the data into training and testing sets
- Feature scaling is applied using StandardScaler to normalize the input features and improve model performance
- The parameter class_weight = 'balanced' is used to handle class imbalance in the dataset
- This ensures that the model gives appropriate importance to both fraud (minority class) and normal (majority class) transactions
- The model learns the relationship between input features (TransactionAmount, Hour, High_Amount) and the target variable (FraudFlag)

Prediction

- After training, the model is used to predict outcomes on the test dataset
- This generates predicted labels for each transaction
- Each transaction is classified into:
  - **0 → Normal Transaction**
  - **1 → Fraudulent Transaction**
- These predictions are compared with actual values to evaluate model performance

9. MODEL EVALUATION

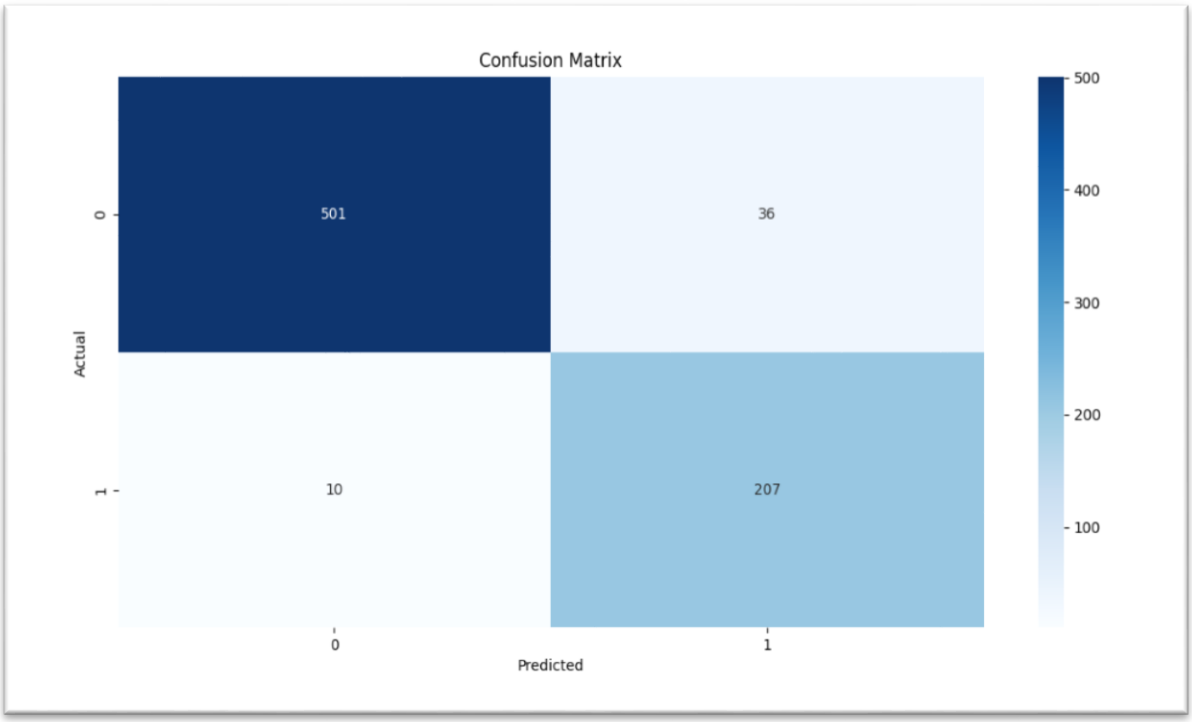
The performance of the model is evaluated using several metrics to understand how effectively it classifies fraudulent and normal transactions.

❖ Accuracy Interpretation

- The model achieved an accuracy of 93.89%
- This indicates that the model correctly classified the majority of transactions as either fraud or normal
- A high accuracy shows good overall performance of the model

❖ Confusion Matrix

- True Negatives (TN) = 501 → Normal transactions correctly identified
- False Positives (FP) = 36 → Normal transactions incorrectly classified as fraud
- False Negatives (FN) = 10 → Fraud transactions missed by the model
- True Positives (TP) = 207 → Fraud transactions correctly identified



❖ **Precision Interpretation**

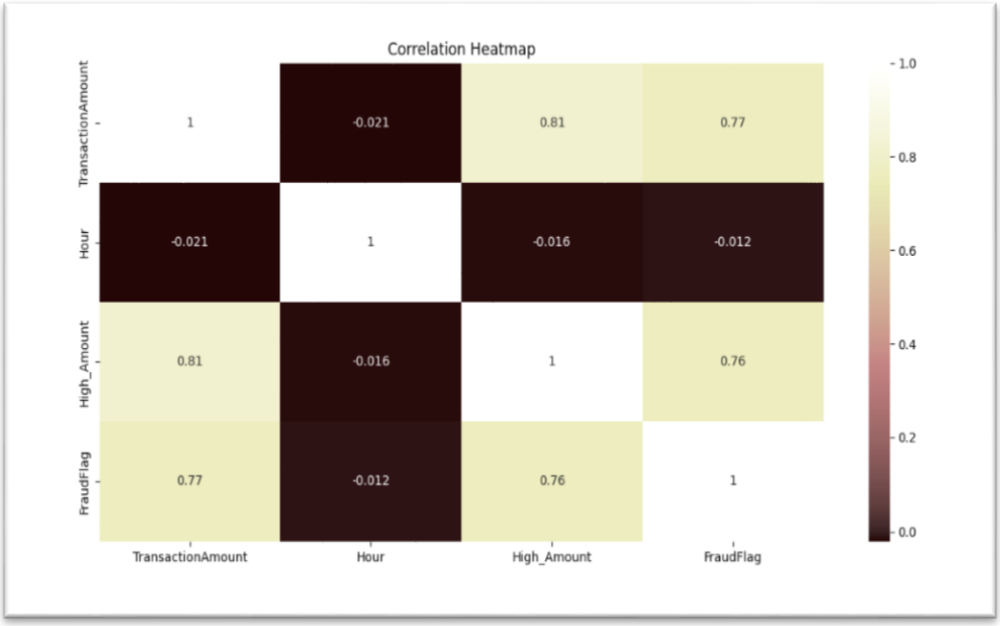
- Precision for fraud class (1) = 0.85
- This means 85% of predicted fraud transactions are actually fraud
- Indicates moderate false alarms

❖ **Recall Interpretation**

- Recall for fraud class (1) = 0.95
- This means 95% of actual fraud transactions are correctly detected
- Very good performance in identifying fraud

❖ **F1-Score Interpretation**

- **F1-score for fraud class = 0.90**
- **Shows a good balance between precision and recall**



10. SQL-Based Analysis

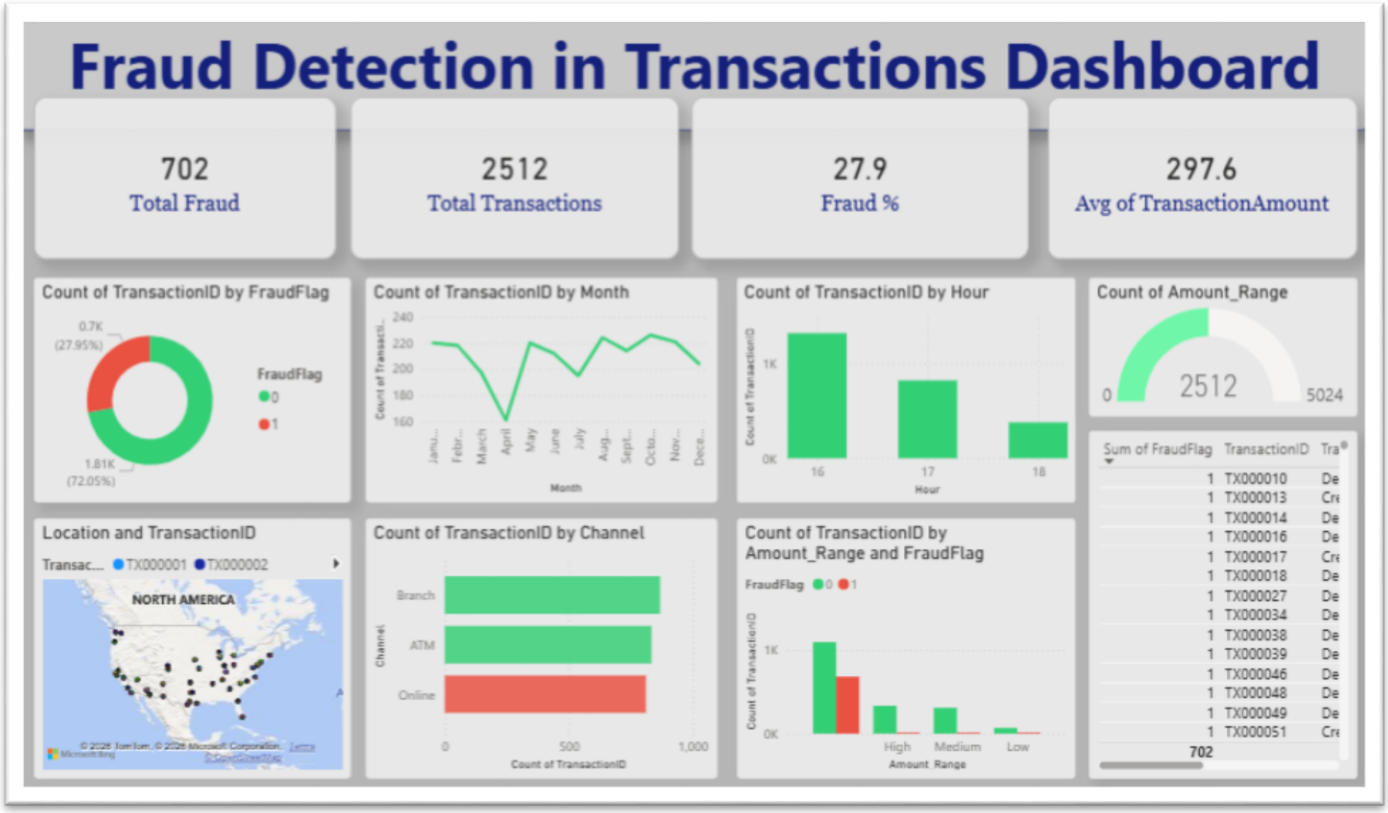
SQL is used in this project to store and analyze transaction data efficiently in a MySQL database. Queries using operations like SELECT, WHERE, GROUP BY, and aggregate functions are performed to calculate total transactions, fraud count, and fraud percentage.

Filtering and aggregation techniques help in identifying high-risk transactions based on factors such as transaction amount, login attempts, and time. SQL also supports pattern analysis across different attributes, enabling better understanding of fraud behavior.

Overall, SQL-based analysis helps in extracting meaningful insights and supports the fraud detection process.

11. DASHBOARD AND VISUALIZATION

❖ **POWERBI**



The dashboard is developed using Power BI Desktop to provide an interactive visualization of transaction data and fraud patterns. It displays key metrics such as total transactions, total fraud cases, fraud percentage, and average transaction amount. Various charts like donut, bar, line, and map visuals are used to analyze fraud distribution, transaction trends over time, high-risk transaction amounts, and location-based activity. The dashboard also includes channel-wise analysis and a detailed transaction table for deeper insights. Overall, it helps in identifying suspicious patterns and supports effective fraud detection and decision-making.

Business Insights :

- Fraud transactions are mostly associated with high transaction amounts
- Suspicious activities are observed during unusual hours (early morning/night)
- Certain transaction channels (Online/ATM) show higher fraud risk
- Specific locations indicate higher concentration of fraudulent activities
- Increased login attempts are linked to potential fraud behavior
- The model achieves high recall, ensuring most fraud cases are detected
- Some false positives exist, but they help in reducing missed fraud cases
- Data-driven insights help organizations improve security and reduce financial losses

Conclusions:

In this project, transaction data was successfully analyzed to detect fraudulent activities using data analysis and machine learning techniques. Feature engineering and exploratory data analysis helped in identifying important patterns such as high transaction amounts, unusual transaction timings, and suspicious user behavior. A Logistic Regression model was implemented to classify transactions as fraudulent or normal, achieving good performance with high accuracy and recall. The developed dashboard provides clear visualization of fraud patterns, enabling better understanding and decision-making. Overall, this project demonstrates how data-driven approaches can effectively identify fraud and help organizations enhance security and minimize financial risks.

Future Scope

- Implement advanced machine learning models like Decision Trees or ensemble methods for improved accuracy
- Develop a real-time fraud detection system for instant monitoring of transactions
- Use larger and more diverse datasets to enhance model performance
- Integrate the system with banking applications for practical implementation
- Apply deep learning techniques for more complex fraud pattern detection
- Improve model performance by handling data imbalance more effectively

References

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- Pandas Documentation
- NumPy Documentation
- Scikit-Learn Documentation
- Matplotlib and Seaborn Documentation
- MySQL Official Documentation
- Power BI Official Documentation